

Hints & Solutions

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Q1 (1) - Single Correct

$$\because A^2 = I \Rightarrow A \cdot A = I$$

$$\text{Also, } A \cdot A^{-1} = I$$

$$\text{That means } A \cdot A = A \cdot A^{-1} \quad A \dots (i)$$

$$\text{As } |A| = (12 - 16) - 1(-12 + 9)$$

$$= -4 + 3$$

$$= -1 \neq 0$$

$$\text{Hence, } A^{-1} \text{ exists.}$$

On multiplying Eq. (i) by A^{-1} , we get

$$A^{-1} \cdot A \cdot A = A^{-1} \cdot A \cdot A^{-1}$$

$$\Rightarrow (A^{-1} \cdot A) \cdot A = (A^{-1} \cdot A) \cdot A^{-1}$$

$$\Rightarrow A = A^{-1}$$

$$\text{Now, } \left(\frac{A}{2}\right)^{-1} = \frac{A^{-1}}{2^{-1}}$$

$$= 2 A^{-1}$$

$$= 2A \quad (\because A^{-1} = A)$$

Q2 (4) - Single Correct

$$\text{Tr}(A + B) = \text{Tr}(A) + \text{Tr}(B)$$

$$\text{Tr}(I_3) = 3$$

$$\text{Tr}(KA) = K \text{Tr}(A)$$

$$\text{But } \text{Tr}(A^2) \neq [\text{Tr}(A)]^2$$

Q3 (2) - Single Correct

$$\Delta_1 = \Delta_0^2, \text{ as } |\text{Cofactors of } A_{n \times n}| = |A|^{n-1}$$

$$\text{Similarly, } \Delta_2 = |\text{Cofactors of } \Delta_1| = \Delta_1^2 = \Delta_0^4 = \Delta_0^{2^2}$$

$$\therefore \Delta_n = |\text{Cofactors of } \Delta_{n-1}| = \Delta_{n-1}^2 = \Delta_0^{2^n}$$

Q4 (3) - Single Correct

$$\because A^2 = 2A - I$$

$$\text{Now, } A^3 = A \cdot A^2$$

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$$= A(2A - I)$$

$$= 2A^2 - A$$

$$= 2(2A - I) - A$$

$$= 4A - A - 2I$$

$$= 3A - 2I$$

$$\text{Now, } A^4 = A^2 \cdot A^2$$

$$= (2A - I)(2A - I)$$

$$= 4A^2 - 2A - 2A + I$$

$$= 4A^2 - 4A + I$$

$$= 4(2A - I) - 4A + I$$

$$= 8A - 4I - 4A + I$$

$$= 4A - 3I$$

$$\text{Hence, } A^n = nA - (n-1)I$$

Q5 (1) - Single Correct

$$\text{Given, } A = \begin{bmatrix} 2 & 1 \\ 4 & 1 \end{bmatrix}$$

$$B = \begin{bmatrix} 3 & 4 \\ 2 & 3 \end{bmatrix}$$

$$C = \begin{bmatrix} 3 & -4 \\ -2 & 3 \end{bmatrix}$$

$$BC = \begin{bmatrix} 3 & 4 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 3 & -4 \\ -2 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 9 & -8 & -12 & 12 \\ 6 & -6 & -8 & 9 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

$$\text{Now, } Tr(A) + Tr\left(\frac{ABC}{2}\right) + Tr\left(\frac{A(BC)^2}{4}\right) + Tr\left(\frac{A(BC)^3}{8}\right) + \dots$$

$$= 3 + \frac{3}{2} + \frac{3}{4} + \frac{3}{8} + \dots$$

$$= 3 \left[1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots \right]$$

$$= \frac{3}{1 - \frac{1}{2}} = 6$$

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Q6 (1) - Single Correct

$$\begin{vmatrix} 1+a+x & a+y & a+z \\ b+x & 1+b+y & b+z \\ c+x & c+y & 1+c+z \end{vmatrix} \\
 = \begin{vmatrix} 1+a+x & b+x & c+x \\ a+y & 1+b+y & c+y \\ a+z & b+z & 1+c+z \end{vmatrix} \\
 = \begin{vmatrix} 1+a+b+c+3x & b+x & c+x \\ 1+a+b+c+3y & 1+b+y & c+y \\ 1+a+b+c+3z & b+z & 1+c+z \end{vmatrix} \\
 = (1+a+b+c) \begin{vmatrix} 1 & b+x & c+x \\ 1 & 1+b+y & c+y \\ 1 & b+z & 1+c+z \end{vmatrix} + 3 \begin{vmatrix} x & b+x & c+x \\ y & 1+b+y & c+y \\ z & b+z & 1+c+z \end{vmatrix} \\
 = (1+a+b+c) \begin{vmatrix} 1 & b+x & c+x \\ 1 & b+z & 1+c+z \\ 0 & y+1-x & y-x \end{vmatrix} + 3 \begin{vmatrix} x & b+x & c+x \\ y & 1+b & c \\ z & b & 1+c \end{vmatrix} \\
 = (1+a+b+c)[(y+1-x)(1+z-x) - (z-x)(y-x)] \\
 + 3[x\{(1+b)(1+c) - bc\} + b\{zc - y(1+c)\} + c\{by - z(1+b)\}] \\
 = (1+a+b+c)[y + yz - yx + 1 + z - x - x - zx + x^2 \\
 - yz + zx + xy - x^2] + 3[x(1+b+c) + b\{zc - y - yc\} + c\{by - z - zb\}] \\
 = (1+a+b+c)[1 + y + z - zx] + 3[x + bx + cx + bcz - by - bcy + bcy - cz - bcz] \\
 = (1+a+b+c)[1 + y + z - 2x] + 3(x + bx + cx - by - bcy + bcy - cz) \\
 = (1+a+b+c)[1 + y + z + x] - 3x(1+a+b+c) + 3(x + bx + cx - by - cz) \\
 = (1+a+b+c)[1 + x + y + z] - 3x - 3ax - 3bx - 3cx + 3x + 3bx + 3cx - 3by - 3cz \\
 = (1+a+b+c)(1+x+y+z) - 3(ax + by + cz)$$

Q7 (4) - Single Correct

$$\text{Given, } A = \begin{bmatrix} a+b & ab & 0 \\ 1 & a+b & ab \\ 0 & 1 & a+b \end{bmatrix} \\
 \therefore |A| = (a+b)[a^2 + b^2 + 2ab - ab] + ab(-a-b) \\
 = (a+b)(a^2 + b^2 + ab) - a^2b - ab^2 \\
 = (a^2 + b^2)(a+b) + a^2b + ab^2 - a^2b - ab^2 \\
 = (a^2 + b^2)(a+b)$$

Q8 (2) - Single Correct

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We have, $|D_2| = \begin{vmatrix} 0 & 1 \\ 2 & 0 \end{vmatrix} = -2$

and $|D_3| = \begin{vmatrix} 0 & 3 & 1 \\ 3 & 0 & 3 \\ 3 & 3 & 0 \end{vmatrix}$
 $= 0 - 3(-9) + 1(9) = 36$

$\therefore |D_2| + |D_3| = 34$

Q9 (4) - Single Correct

$$A = \begin{bmatrix} 1 & 2r-1 \\ 0 & 1 \end{bmatrix}$$

$$\prod_{r=1}^{50} \begin{bmatrix} 1 & 2r-1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & \sum_{r=1}^{50} (2r-1) \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 2500 \\ 0 & 1 \end{bmatrix}$$

Q10 (3) - Single Correct

Since, $(2n+1)^2 - (2m+1)^2 = 4(m+n+1)(n-m)$

$\therefore$ Multiple of 8

Q11 (2) - Single Correct

Here, $(AB)(AB) = A(BA)B = A(A^2B)B = A^3B^2$

Now, $(AB)(AB)(AB) = (A^3B^2)AB$

$= A^3B^2AB = A^3B(BA)B$

$= A^3B(A^2B)B = A^3(BA) \cdot AB^2$

$= A^3(A^2B) \cdot AB$

$= A^5BAB^2 = A^5 \cdot A^2B \cdot B^2 = A^7 \cdot B^3$

So, $(AB)^n = A^{2n-1} \cdot B^n$

$\therefore (AB)^{10} = A^{2^{10}-1} \cdot B^{10} \Rightarrow k = 2^{10} - 1 = 1023$

Q12 (4) - Single Correct

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For non-trivial solution,

$$\begin{vmatrix} \alpha & \beta & \gamma \\ \beta & \gamma & \alpha \\ \gamma & \alpha & \beta \end{vmatrix} = 0 \Rightarrow \alpha^3 + \beta^3 + \gamma^3 - 3\alpha\beta\gamma = 0$$

$$\Rightarrow (\alpha + \beta + \gamma)(\alpha^2 + \beta^2 + \gamma^2 - \alpha\beta - \beta\gamma - \alpha\gamma) = 0$$

$$\Rightarrow (-a)[(\alpha + \beta + \gamma)^2 - 3(\alpha\beta + \beta\gamma + \gamma\alpha)] = 0$$

$$\Rightarrow (-a)[a^2 - 3b] = 0$$

$$\Rightarrow a^3 - 3ab = 0$$

$$\text{or } a^3 = 3ab$$

$$\therefore a^3 + 6c = 3ab + 6c$$

Q13 (4) - Single Correct

Given,

$$P = \begin{vmatrix} c-a & c-b & ab-c^2 \\ a-b & a-c & bc-a^2 \\ b-c & b-a & ca-b^2 \end{vmatrix} = \begin{vmatrix} a & b & 1 \\ b & c & 1 \\ c & a & 1 \end{vmatrix}^2 = (2010)^2$$

$$= (2 \times 3 \times 5 \times 67)^2 = 2^2 \times 3^2 \times 5^2 \times 67^2$$

$$\therefore \text{Number of divisors} = (2+1)(2+1)(2+1)(2+1) = 81$$

Q14 (4) - Single Correct

$$\left. \begin{matrix} AB = A \\ BA = B \end{matrix} \right\} \Rightarrow \left. \begin{matrix} A^2 = A \\ B^2 = B \end{matrix} \right\} \Rightarrow \left. \begin{matrix} A^n = A \\ B^n = B \end{matrix} \right\}$$

$$\text{and } (A+B)^2 = A^2 + B^2 + AB + BA$$

$$= A + B + A + B = 2(A+B)$$

$$\therefore (A^{2010} + B^{2010})^{2011} = (A+B)^{2011} = 2^{2010}(A+B)$$

Q15 (3) - Single Correct

$$\text{Let } A = (a_{ij})_{3 \times 3}$$

$$\text{and } X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$\text{Now, } X^T A X = 0$$

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$$\Rightarrow \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

$$\Rightarrow a_{11}x^2 + a_{22}y^2 + a_{33}z^2 + (a_{12} + a_{21})xy + (a_{13} + a_{31})xz + (a_{23} + a_{32})yz = 0$$

$$a_{23} = -a_{32} \text{ and } a_{11} = a_{22} = a_{33} = 0$$

$$\begin{aligned} \text{Hence, } a_{21} + a_{31} + a_{32} &= -(a_{12} + a_{13} + a_{23}) \\ &= -(2008 + 2010 - 2012) = -2006 \end{aligned}$$

Q16 (1, 2, 3) - Multiple Correct

Here, $A^2 = A$ and $B^2 = B$, also $AB \pm BA = 0$

$$\therefore A^2 - B^2 = A - B$$

$$\Rightarrow (A - B)[(A + B) - 1] = 0$$

Either $A - B = 0$ or $A + B = 1 \dots (i)$

$$\text{Now, } (A - B)^2 = A^2 + B^2 + AB - BA$$

$$= A^2 + B^2 + 0$$

$$\Rightarrow (A - B)^2 = A + B$$

$$\therefore |A - B|^2 = |A + B|, \quad [\text{From Eq. (i), } |A + B| = 1]$$

$$\Rightarrow |A - B|^2 = 1 \text{ or } |A - B| = \pm 1 \dots (ii)$$

Again, from Eq. (i), $A - B = 0$

$$\Rightarrow |A - B| = 0 \dots (iii)$$

$$\therefore |A - B| = 0, -1, 1$$

Q17 (1, 2) - Multiple Correct

$$AA^T = A^T A = I$$

$$\Rightarrow (A^T)^T A^T = A^T (A^T)^T = I$$

Hence, A^T is orthogonal.

$$\therefore AA^T = I$$

$$\Rightarrow (AA^T)^{-1} = I^{-1}$$

$$\Rightarrow (A^T)^{-1} A^{-1} = I \Rightarrow (A^{-1})^T A^{-1} = I$$

$$\text{Similarly, } A^{-1} (A^{-1})^T = I$$

Hence, A^{-1} is orthogonal.

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Q18 (4) - Multiple Correct

The transpose of a diagonal matrix is itself. Also, the multiplication of A and D will be always commutative, where A and D are matrices of order 3×3 and D is a diagonal matrix.

Q19 (1, 2, 3, 4) - Multiple Correct

If $|A| \neq 0$

$\Rightarrow A^{-1}$ exists.

$$AB = 0$$

$$\Rightarrow A^{-1}(AB) = 0$$

$$\Rightarrow B = 0$$

again if $|B| \neq 0$

$\Rightarrow B^T$ exists.

$$AB = 0$$

$$\Rightarrow (AB)B^{-1} = 0$$

$$\Rightarrow A = 0$$

$$\therefore AB = 0$$

$$\Rightarrow |AB| = 0$$

$$\Rightarrow |A||B| = 0$$

If $|A| \neq 0$

Then $|B| = 0$

and $|B| \neq 0$

Then $|A| = 0$

Q20 (1, 2, 3) - Multiple Correct

$$\text{Here, } B = A^{2^n} = A^{2^{n-1} \cdot 2} = (A^2)^{2^{n-1}}$$

$$= (A^{-1})^{2^{n-1}} \quad [u \text{ s.t. } A^2 = A^{-1}]$$

$$= (A^{-1})^{2^{n-2} \cdot 2^1}$$

$$= \left((A^{-1})^2 \right)^{2^{n-2}}$$

$$= \left((A^2)^{-1} \right)^{2^{n-2}}$$

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$$= \left((A^{-1})^{-1} \right)^{2^{n-2}}$$

$$= A^{2^{n-2}}$$

$$= C$$

$\therefore B = C$ Option (3) is correct.

$\Rightarrow |B - C| = 0$, option (1) is correct. $\Rightarrow (B + C)(B - C) = 0$, option (2) is correct.

Q21 (1, 3) - Multiple Correct

Given, $f(x) = \begin{vmatrix} p + \sin x & p + d + \sin x & -2d + \sin x \\ p + d + \sin x & p + 2d + \sin x & -1 + \sin x \\ p + 2d + \sin x & p + 3d + \sin x & 2d + \sin x \end{vmatrix}$

On applying $C_1 \rightarrow C_1 - C_2$

$$= \begin{vmatrix} -d & p + d + \sin x & -2d + \sin x \\ -d & p + 2d + \sin x & -1 + \sin x \\ -d & p + 3d + \sin x & 2d + \sin x \end{vmatrix}$$

$$= -d \begin{vmatrix} 1 & p + d + \sin x & -2d + \sin x \\ 1 & p + 2d + \sin x & -1 + \sin x \\ 1 & p + 3d + \sin x & 2d + \sin x \end{vmatrix}$$

On applying $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - R_1$

$$= -d \begin{vmatrix} 1 & p + d + \sin x & -2d + \sin x \\ 0 & d & -1 + 2d \\ 0 & 2d & 4d \end{vmatrix}$$

$$= -d [4d^2 + 2d - 4d^2] = -d \times 2d = -2d^2$$

$$\therefore \int_0^2 f(x) dx = -4$$

$$\Rightarrow -2d^2 \int_0^2 dx = -4$$

$$\Rightarrow 4d^2 = +4$$

$$\Rightarrow d^2 = 1$$

$$\therefore d = \pm 1$$

Q22 (4) - Multiple Correct

If $\alpha = -2$

Then, $-x - 1 = 0$

and $x + y - 1 = 0$

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$$x = -1, y = 2$$

$$\text{If } \alpha = 1$$

$$\text{Then, } 8x + 27y - 64 = 0$$

$$2x + 3y - 4 = 0$$

$$x + y - 1 = 0$$

$$[A : B] = \left[\begin{array}{ccc|c} 1 & 1 & 0 & 1 \\ 2 & 3 & 0 & 4 \\ 8 & 27 & 0 & 64 \end{array} \right]$$

$$\text{On applying } R_3 \rightarrow R_3 - 8R_1 \text{ and } R_2 \rightarrow R_2 - 2R_1$$

$$= \left[\begin{array}{ccc|c} 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 19 & 0 & 56 \end{array} \right]$$

$$\text{On applying } R_3 \rightarrow R_3 - 19R_2$$

$$= \left[\begin{array}{ccc|c} 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 16 \end{array} \right]$$

$$\text{So, Rank}(A) \neq \text{Rank}(AB)$$

So, the system of equations is inconsistent. Similarly, for $\alpha = 0$ and $\alpha = -3$ We can prove that the given system of equations is inconsistent.

Q23 (1, 4) - Multiple Correct

$$\Delta = \begin{vmatrix} x & 2y - z & -z \\ y & 2x - z & -z \\ z & 2y - z & 2x - 2y + 2 \end{vmatrix}$$

$$\text{On applying } R_1 \rightarrow R_1 - R_2$$

$$= \begin{vmatrix} x - y & 2(y - x) & 0 \\ y & 2x - z & -z \\ z & 2y - z & 2x - 2y + z \end{vmatrix}$$

$$= (x - y) \begin{vmatrix} 1 & -2 & 0 \\ y & 2x - z & -z \\ z & 2y - z & 2x - 2y + z \end{vmatrix}$$

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On applying $C_2 \rightarrow C_2 + 2C_1$

$$\begin{aligned} & (x-y) \begin{vmatrix} 1 & 0 & 0 \\ y & 2x+2y-z & -z \\ z & 2y+z & 2x-2y+z \end{vmatrix} \\ &= (x-y)[(2x+2y-z)(2x-2y+z) + z(2y+z)] \\ &= (x-y)[4x^2 - (2y-z)^2 + 2yz + z^2] \\ &= (x-y)[4x^2 - 4y^2 + 4yz + 2yz] \\ &= (x-y)(4x^2 - 4y^2 + 6yz) \end{aligned}$$

Q24 (1, 2, 3) - Multiple Correct

$$\Delta = \begin{vmatrix} -x & a & b \\ b & -x & a \\ a & b & -x \end{vmatrix}$$

On applying $R_1 \rightarrow R_1 + R_2 + R_3$

$$\begin{vmatrix} a+b-x & a+b-x & a+b-x \\ b & -x & a \\ a & b & -x \end{vmatrix} = (a+b-x) \begin{vmatrix} 1 & 1 & 1 \\ b & -x & a \\ a & b & -x \end{vmatrix}$$

On applying $C_2 \rightarrow C_2 - C_1$ and $C_3 \rightarrow C_3 - C_1$

$$\begin{aligned} &= (a+b-x) \begin{vmatrix} 1 & 0 & 0 \\ b & -x-b & a-b \\ a & b-a & -x-a \end{vmatrix} \\ &= (a+b-x)[(-x-b)(-x-a) - (a-b)(b-a)] \\ &= (a+b-x)[x^2 + (a+b)x + ab + a^2 + b^2 - 2ab] \\ &= (a+b-x)(x^2 + (a+b)x + a^2 + b^2 - ab) \end{aligned}$$

Q25 (2, 4) - Multiple Correct

$$\begin{vmatrix} b & c & b\alpha + c \\ c & d & c\alpha + d \\ b\alpha + c & c\alpha + d & a\alpha^3 - c\alpha \end{vmatrix} = 0$$

On applying $R_3 \rightarrow R_3 - \alpha R_1$

$$\begin{vmatrix} b & c & b\alpha + c \\ c & d & c\alpha + d \\ c & d & a\alpha^3 - (b+c)\alpha - \alpha c \end{vmatrix} = 0$$

On applying $R_3 \rightarrow R_3 - R_2$

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$$\Rightarrow \begin{vmatrix} b & c & b\alpha + c \\ c & d & c\alpha + d \\ 0 & 0 & a\alpha^3 - (b+c)\alpha - 2c\alpha - d \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} 0 & 0 & a\alpha^3 - (b+c)\alpha - 2c\alpha - d \\ c & d & c\alpha + d \\ b & c & b\alpha + c \end{vmatrix} = 0$$

$$\Rightarrow [a\alpha^3 - (b+c)\alpha - 2c\alpha - d] (c^2 - bd) = 0$$

$\therefore b, c, d$ are in GP and α is a root of $ax^3 - bx^2 - 3cx - d = 0$

Q26 (1, 2) - Multiple Correct

$$\Delta = \begin{vmatrix} a & a^2 & 0 \\ 1 & 2a+b & (a+b)^2 \\ 0 & 1 & 2a+3b \end{vmatrix}$$

On applying $R_2 \rightarrow aR_2 - R_1$

$$= \frac{1}{a} \begin{vmatrix} a & a^2 & 0 \\ 0 & a^2 + ab & a(a+b)^2 \\ 0 & 1 & 2a+3b \end{vmatrix}$$

$$= \{ (a^2 + ab)(2a+3b) - a(a+b)^2 \}$$

$$= a \{ (a+b)(2a+3b) - (a+b)^2 \}$$

$$= a(a+b) \{ 2a+3b - a - b \}$$

$$\therefore = a(a+b)(a+2b)$$

Q27 (3, 4) - Multiple Correct

$$A = \begin{bmatrix} \alpha & 2 \\ 2 & \alpha \end{bmatrix}$$

$$\text{and } |A^3| = 125$$

$$A^2 = \begin{bmatrix} \alpha & 2 \\ 2 & \alpha \end{bmatrix} \begin{bmatrix} \alpha & 2 \\ 2 & \alpha \end{bmatrix} = \begin{bmatrix} \alpha^2 + 4 & 4\alpha \\ 4\alpha & 4 + \alpha^2 \end{bmatrix}$$

$$\therefore A^3 = A^2 \cdot A$$

$$= \begin{bmatrix} \alpha^2 + 4 & 4\alpha \\ 4\alpha & \alpha^2 + 4 \end{bmatrix} \begin{bmatrix} \alpha & 2 \\ 2 & \alpha \end{bmatrix}$$

$$= \begin{bmatrix} \alpha^3 + 12\alpha & 6\alpha^2 + 8 \\ 6\alpha^2 + 8 & \alpha^3 + 12\alpha \end{bmatrix}$$

$$\Rightarrow |A|^3 = (\alpha^3 + 12\alpha)^2 - (6\alpha^2 + 8)^2$$

If $\alpha = 3$, then

$$= (27 + 36)^2 - (54 + 8)^2$$

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$$= (63)^2 - (62)^2$$

$$= (63 + 62) \times 1 = 125$$

If $\alpha = -3$

$$\text{Then, } |A|^3 = (-27 - 36)^2 - (54 + 8)^2 = 125$$

Q28 (1) - Paragraph 1

We have, $C^T(A + B)C$

$$\Rightarrow C^T(A + B)(A + B)^{-1}(A - B)$$

$$= C^T(A - B)$$

$$= [(A + B)^{-1}(A - B)]^T(A - B)$$

$$= (A - B)^T((A + B)^T)^{-1}(A - B)$$

$$= (A + B)(A - B)^{-1}(A - B)$$

$$= A + B$$

Q29 (2) - Paragraph 1

We have,

$$C^T(A - B)C$$

$$= [(A + B)^{-1}(A - B)]^T(A - B)C$$

$$= (A - B)^T[(A + B)^T]^{-1}(A - B)C$$

$$= (A + B)(A - B)^{-1}(A - B)C$$

$$= (A + B)C$$

$$= (A + B)(A + B)^{-1}(A - B)$$

$$= A - B$$

Q30 (3) - Paragraph 2

Let A = Skew-symmetric matrix

and B = Scalar matrix

$$\begin{aligned} \therefore n(A \cup B) &= n(A) + n(B) - n(A \cap B) \\ &= 7^3 + 7 \cdot (1) - 1 = 7^3 + 6 \end{aligned}$$

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Q31 (3) - Paragraph 2

Let A = Lower diagonal matrix

B = Upper diagonal matrix

$$\begin{aligned} \therefore n(A \cup B) &= n(A) + n(B) - n(A \cap B) \\ &= 7^6 + 7^6 - 7^3 = 2 \cdot 7^6 - 7^3 = 7^3 (2 \times 7^3 - 1) \end{aligned}$$

Q32 (2) - Paragraph 3

$$\therefore |A - \lambda I| = 0$$

$$\text{If } \lambda = 0$$

$$\text{Then, } |A| = 0$$

Q33 (4) - Paragraph 3

$$\text{Given, } AA' = I$$

$$\text{and } |A| = 1$$

$$\text{Suppose } A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

$$\Rightarrow |A - I| = 0$$

$$\text{If } A = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$\Rightarrow A - I = \begin{bmatrix} -2 & 0 \\ 0 & -2 \end{bmatrix}$$

$$\therefore |A - I| = 4$$

Q34 (3) - Paragraph 4

$$A^2 = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & -1 \\ 3 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & -1 \\ 3 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 4 & 3 & 0 \\ -3 & 2 & -2 \\ 6 & 4 & 5 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} 4 & 3 & 0 \\ -3 & 2 & -2 \\ 6 & 4 & 5 \end{bmatrix} \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & -1 \\ 3 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 4 & 11 & 1 \\ -9 & -2 & -7 \\ 21 & 11 & 7 \end{bmatrix}$$

$$A^3 + kA^2 + lA + ml = 0$$

$$\begin{bmatrix} 4 + 4k + l + m & 11 + 3k + 2l & 1 + l \\ -9 - 3k & -2 + 2k + l + m & -7 - 2k - l \\ 21 + 6k + 3l & 11 + 4k & 7 + 5k + l + m \end{bmatrix} = 0$$

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$$\Rightarrow l = -1, k = -3$$

$$\text{and } m = 9$$

Q35 (3) - Paragraph 4

$$\text{So, } (k + m + l) = -3 + 9 - 1 = 5$$

$$\therefore A^3 - 3A^2 - A + 9I = 0$$

$$\Rightarrow A^2 - 3A - 1 + 9A^{-1} = 0$$

$$\Rightarrow -9A^{-1} = A^2 - 3A - 1$$

$$\therefore A^{-1} = \frac{-1}{9}(A^2 - 3A - 1)$$

Q36 (1) - Paragraph 5

A can be written as

$$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

Q37 (2) - Paragraph 5

$$\text{Given, } A \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

For unique solution, $\det(A) \neq 0$

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Case I

$$\det(A) = \begin{vmatrix} 1 & a & b \\ a & 1 & c \\ b & c & 1 \end{vmatrix} = 2abc \neq 0$$

Here, a, b, c is selected from $1, 0, 0$. [no case is possible]

Case II

$$(i) \det(A) = \begin{vmatrix} 1 & a & b \\ a & 0 & c \\ b & c & 0 \end{vmatrix} = 2abc - c^2 \neq 0$$

Here, a, b, c are selected from $1, 1, 0$. [2 cases are possible]

$$(ii) \det(A) = \begin{vmatrix} 0 & a & b \\ a & 1 & c \\ b & c & 0 \end{vmatrix} = 2abc - b^2 \neq 0$$

Here, a, b, c are selected from $1, 1, 0$. [2 cases are possible]

$$(iii) \det(A) = \begin{vmatrix} 0 & a & b \\ a & 0 & c \\ b & c & 1 \end{vmatrix}$$

$$2abc - a^2 \neq 0$$

Here, a, b, c are selected from $1, 1, 0$. [2 cases are possible]

Hence, there are exactly 6 matrices for unique solution.

Q38 (2) - Integer Type

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

Suppose $n = 1$

$$|A - I| = 1 - \lambda$$

$$\Rightarrow \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} = 1 - \lambda$$

$$\Rightarrow -1 = 1 - \lambda$$

$$\therefore \lambda = 2$$

Q39 (6) - Integer Type

$$\therefore P^2 = I - P$$

$$\text{Now, } P^3 = P^2 \cdot P$$

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$$= (I - P)P$$

$$= P - P^2$$

$$= P - (I - P)$$

$$= P - I + P$$

$$= 2P - I$$

$$\text{Now, } P^4 = P^2 \cdot P^2$$

$$= (I - P)(I - P) = I^2 - 2P + P^2$$

$$= I - 2P + I - P$$

$$= 2I - 3P$$

$$\Rightarrow P^5 = P^4 \cdot P$$

$$= (2I - 3P)P = 2P - 3P^2$$

$$= 2P - 3(I - P)$$

$$= 2P + 3P - 3I = 5P - 3I$$

$$\text{Now, } P^6 = P^5 \cdot P$$

$$= (5P - 3I)P$$

$$= 5P^2 - 3P$$

$$= 5(I - P) - 3P$$

$$= 5I - 8P$$

$$\text{Hence, } n = 6$$

Q40 (2) - Integer Type

$$\text{We have, } (I - 0.4A)^{-1} = 1 - \lambda A$$

$$\Rightarrow I = (I - 0.4A)(I - \lambda A)$$

$$\Rightarrow I = I - (\lambda + 0.4A) + 0.4\lambda A^2$$

$$\Rightarrow 0 = -\lambda - 0.4A + 0.4\lambda A$$

$$\Rightarrow 0 = -0.6\lambda A - 0.4A$$

$$\Rightarrow \lambda = \frac{-0.4}{-0.6} = \frac{2}{3}$$

$$\text{Now, } |3\lambda| = \left| 3 \times \left(\frac{2}{3} \right) \right| = 2$$

Q41 (2) - Integer Type

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Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

$A^T = \begin{bmatrix} a & c \\ b & d \end{bmatrix}$

$\therefore A + A^T = 1$

$\Rightarrow \begin{bmatrix} 2a & b+c \\ b+c & 2d \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$\Rightarrow a = d = \frac{1}{2} \text{ and } b = -c$

Hence, $A = \begin{bmatrix} \frac{1}{2} & -c \\ c & \frac{1}{2} \end{bmatrix}$

$\therefore AA^T = 1$
 $\Rightarrow \begin{bmatrix} \frac{1}{2} & -c \\ c & \frac{1}{2} \end{bmatrix} \begin{bmatrix} \frac{1}{2} & c \\ -c & \frac{1}{2} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$\Rightarrow \begin{bmatrix} \frac{1}{4} + c^2 & 0 \\ 0 & c^2 + \frac{1}{4} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$\Rightarrow \frac{1}{4} + c^2 = 1$

$\Rightarrow c^2 = \frac{3}{4}$

$\Rightarrow c = \pm \frac{\sqrt{3}}{2}$

Hence, $A = \begin{bmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix}$

and $A^T = \begin{bmatrix} \frac{1}{2} & -\sqrt{3} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix}$

Q42 (2) - Integer Type

$\therefore AA^T = 1$

$\Rightarrow \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} a & c \\ b & d \end{bmatrix} = 1$

$\Rightarrow \begin{bmatrix} a^2 + b^2 & ac + bd \\ ac + bd & c^2 + d^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$\Rightarrow a^2 = 1, d^2 = 1$

$b = 0 \text{ and } c = 0$

or $b^2 = 1, c^2 = 1$

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$$a = 0 \text{ and } d = 0$$

$$\text{Hence, } A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \text{ or } A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Q43 (7) - Integer Type

$$\text{We have, } A = \begin{bmatrix} 3 & 11 \\ 2 & 8 \end{bmatrix}$$

$$\Rightarrow |A| = 24 - 22 = 2$$

$$\text{Now, } |A^{2011} - 5A^{2010}|$$

$$= |A^{2010}(A - 5)|$$

$$= |A^{2010}| |A - 5|$$

$$= (2)^{2010} \times (-28)$$

$$= -7 \times (2)^{2012}$$

$$\text{Hence, } \lambda = 7$$

Q44 (9) - Integer Type

$$\text{Given, } A^2 = A \text{ and } (I + A)^{10} = 1 + kA, \text{ where } k = 2 \cdot 2^\lambda - 1$$

$$\text{Now, } (I + A)^{10} = 1 + A + \frac{A^2}{2!} + \frac{A^3}{3!} + \dots + \frac{A^{10}}{10!}$$

$$= 1 + A + \frac{A}{2!} + \frac{A}{3!} + \dots + \frac{A}{10!}$$

$$= 1 + A \left(1 + \frac{1}{2!} + \frac{1}{3!} + \dots + \frac{1}{10!} \right)$$

$$= 1 + A(1 + 1)^{10}$$

$$= 1 + A \times (2)^{10} \text{ (given)}$$

$$= 1 + kA$$

$$k = 2^{\lambda+1} - 1$$

$$\Rightarrow k = 2^{10}$$

$$\Rightarrow 2^{\lambda+1} - 1 = 2^{10}$$

$$\Rightarrow 2^{\lambda+1-10} = 1$$

$$\Rightarrow \lambda + 1 - 10 = 0$$

$$\Rightarrow \lambda = 9$$

Q45 (10) - Integer Type

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$$\text{Given, } A(\text{adj } A) = \begin{bmatrix} 10 & 0 \\ 0 & 10 \end{bmatrix}$$

$$\Rightarrow |A(\text{adj } A)| = 100$$

$$\Rightarrow |A| |\text{adj } A| = 100$$

$$\Rightarrow |A| |A|^{2-1} = 100$$

$$\Rightarrow |A| \cdot |A| = 100 \Rightarrow |A|^2 = 100$$

$$\therefore |A| = 10$$

Q46 (4) - Integer Type

$$\text{Given, } A = \begin{bmatrix} 2k-1 & 2\sqrt{k} & 2\sqrt{k} \\ 2\sqrt{k} & 1 & -2k \\ -2\sqrt{k} & 2k & -1 \end{bmatrix}$$

$$\Rightarrow |A| = (2K-1)[-1+4k^2] + 2\sqrt{k}[4K\sqrt{k}+2\sqrt{k}] + 2\sqrt{k}[4k\sqrt{k}+2\sqrt{k}]$$

$$= (2k-1)(-1+4k^2) + 4k(2k+1) + 4k[2k+1]$$

$$= -2k + 8k^3 + 1 - 4k^2 + 16k^2 + 8k$$

$$= 8k^3 + 12k^2 + 6k + 1 = (2k+1)^3$$

$$\text{and } B = \begin{bmatrix} 0 & 2k-1 & \sqrt{k} \\ 1-2k & 0 & 2\sqrt{k} \\ -\sqrt{k} & -2\sqrt{k} & 0 \end{bmatrix}$$

$$\Rightarrow |B| = 0$$

[$\because B$ is a skew-symmetric matrix of odd order]

$$\text{Now, } |\text{adj } A| + |\text{adj } B| = |A|^{3-1} + |B|^{3-1}$$

$$= |A|^2 + |B|^2 = (2k+1)^6$$

$$\text{Given, } |\text{adj } A| + |\text{adj } B| = 10^6$$

$$\Rightarrow (2k+1)^6 = 10^6$$

$$\Rightarrow 2k+1 = 10$$

$$\Rightarrow 2k = 9 \Rightarrow k = 4.5$$

$$[k] = 4$$

Q47 (2) - Integer Type

$$|2A^9 B^{-1}| = 2^2 |A^9 B^{-1}|$$

[$\because A^9 B^{-1}$ is a matrix of order 2×2]

Hints & Solutions

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$$= 4 |A^9| |B^{-1}|$$

$$= \frac{4|A|^9}{|B|} = \frac{4 \times (-1)^9}{2}$$

$$= -2$$

Modulus of $|2A^9B^{-1}|$ is 2

Q48 (6) - Integer Type

$$f(x) = \begin{vmatrix} 1 + \sin^2 x & \cos^2 x & 4 \sin 2x \\ \sin^2 x & 1 + \cos^2 x & 4 \sin 2x \\ \sin^2 x & \cos^2 x & 1 + 4 \sin 2x \end{vmatrix}$$

On applying $C_1 \rightarrow C_1 + C_2 + C_3$

$$= \begin{vmatrix} 2 + 4 \sin 2x & \cos^2 x & 4 \sin 2x \\ 2 + 4 \sin 2x & 1 + \cos^2 x & 4 \sin 2x \\ 2 + 4 \sin 2x & \cos^2 x & 1 + 4 \sin 2x \end{vmatrix} \left[\because \sin^2 x + \cos^2 x = 1 \right]$$

$$= (2 + 4 \sin 2x) \begin{vmatrix} 1 & \cos^2 x & 4 \sin 2x \\ 1 & 1 + \cos^2 x & 4 \sin 2x \\ 1 & \cos^2 x & 1 + 4 \sin 2x \end{vmatrix}$$

On applying $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - R_1$

$$= (2 + 4 \sin 2x) \begin{vmatrix} 1 & \cos^2 x & 4 \sin 2x \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

$$= (2 + 4 \sin 2x)$$

Hence, maximum value of $f(x)$ is 6.

Q49 (4) - Integer Type

$$\therefore \frac{5}{3} |A^2 B C^{-1}| = \frac{5}{3} |A \cdot A \cdot B \cdot C^{-1}|$$

$$= \frac{5}{3} |A| |A| |B| |C^{-1}|$$

$$= \frac{5}{3} |A|^2 |B| \times \frac{1}{|C|}$$

$$= \frac{5}{3} \times 4 \times 3 \times \frac{1}{5}$$

$$= 4$$